Pop Quizzes

Quiz 1

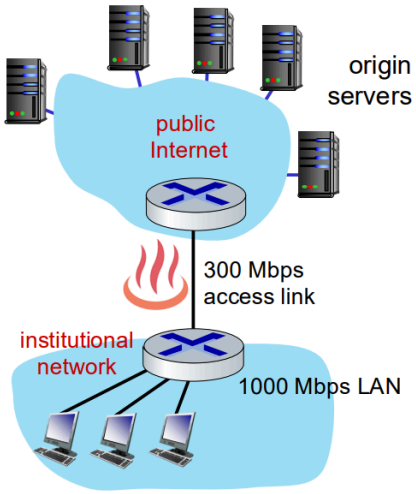
Question?

Scenario:

- access link rate: 300 Mbps
- LAN speed: 1000 Mbps
- RTT from institutional router to server: 1 sec
- web object size: 500K bits
- average request rate from browsers to origin servers: 60/sec
  - avg data rate to browsers: ??

Performance:

- access link utilization = ??
- LAN utilization = ??
- (minimum) end-end delay = ??



Question 1

www.mst.edu uses which transport layer protocol?

- [x] TCP
- Both UDP or TCP
- None of Them
- UDP

Question 2

Look at the given scenario the Whiteboard. What is the LAN utilization (%)?

- 2
- 10
- [x] 3

- 5

Question 3

Look at the given scenario the Whiteboard. What is the minimum end-to-end delay (sec)?

- 2
- [x] 1
- 1.5
- 0.5

Question 4

Look at the given scenario the Whiteboard. What is the access link utilization (%)?

- [x] 10
- 8
- 97
- 15

Question 5

Suppose the server-to-client HTTP RESPONSE message is the following:

```
HTTP/1.1 404 Not Found
Date: Thu, 05 Sep 2024 23:14:55 +0000
Server: Apache/2.2.3 (CentOS)
Content-Length: 2997
Keep-Alive: timeout=42, max=100
Connection: Keep-alive
Content-type: text/html
```

The server was able to send the document successfully.

- True
- [x] False

Quiz 2

Question 1

For the given scenario on the whiteboard, what is the minimum client download time (seconds)?
909.09

Question 2

For the given scenario on the whiteboard, what is the time needed to distribute this file from the central server to the 10 peers using the client-server model (Seconds)?
1,639.34

Question 3

For the given scenario on the whiteboard, in the client/server paradigm, how much data (Gbits) upload is the server responsible for?
100

Question 4

For the given scenario on the whiteboard, what is the time needed to distribute this file from the central server to the 10 peers using the P2P model (Seconds)?
909.09

Pop Quiz 3

Question 1

- ☐ When an IP datagram arrives at a host, which process is responsible for delivering the datagram to the correct transport-layer protocol?
- ☐ Multiplexing

- [ ] Application-layer Addressing
- [x] Demultiplexing
- [ ] IP routing

Question 2

What role does the source port number in a segment header play in multiplexing?

- [ ] Identifies the destination application
- [ ] Routes the segment to the correct IP address
- [x] Identifies the sending application
- [ ] Ensures data integrity

Question 3

The transport layer is a logical communication mechanism between application processes running on different host

- [x] True
- [ ] False

Question 4

What is the primary purpose of the UDP checksum?

- [ ] To ensure the reliability of data transmission by requesting retransmission of lost packets
- [x] To verify the integrity of the data in the UDP segment and detect any errors during transmission
- [ ] To manage flow control between sender and receiver
- [ ] To establish a connection between the sender and receiver before data transmission

Pop Quiz 5

Question 1

Which routing protocol is used for inter-AS routing on the Internet?

- [ ] EIGRP
- [ ] RIP
- [ ] OSPF
- [x] BGP

Question 2

Which of the following best describes an Autonomous System (AS)?

- [ ] A subnet that cannot connect to the Internet
- [ ] A router that performs only link-state operations
- [x] A collection of networks under single administrative control
- [ ] A router within a single LAN

Question 3

In OSPF hierarchical design, link-state advertisements are flooded across the entire AS, including all areas.

- [ ] True
- [x] False

Question 4

Hot-potato routing means:

- [ ] Forwarding traffic only to customers
- [ ] Ignoring routing policies and sending traffic at random
- [x] Choosing the nearest exit point from the AS to forward traffic
- [ ] Selecting the route with the minimum AS-PATH length